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Hepatic parenchymal hypoattenuation in dogs with diabetes mellitus on computed tomography consistent with hepatic steatosis

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Abstract

Hypoattenuation of the liver, consistent with hepatic steatosis or lipidosis, has been reported in veterinary patients. In people, measuring CT hepatic attenuation is diagnostic for hepatic steatosis, and hypoattenuation of the liver is defined as absolute if less than 40 HU or relative if the liver is 10 HU less than the spleen. The purpose of this study is to describe hepatic parenchymal attenuation in dogs with diabetes mellitus with or without diabetic ketosis (DK) or diabetic ketoacidosis (DKA), using the above categorization for absolute and relative hypoattenuation, as with humans. We hypothesized dogs with DK or DKA were more likely to have hypoattenuating livers. Twenty-seven diabetic dogs were included; fifteen were categorized in Group 1 as without DK or DKA, six in Group 2 as DK, and six in Group 3 as DKA. In Group 3, four of six dogs had absolute and relative hypoattenuating livers. Three of these were visually hypoattenuating to the vasculature, with one having negative attenuation and a histopathologic diagnosis of severe hepatic lipidosis. In Group 2, four of six dogs had relative hypoattenuating livers. In Group 1, only one of 15 dogs had a relatively hypoattenuating liver. Groups 2 and 3 had significantly lower absolute liver attenuation than Group 1. Presumed hepatic steatosis was present on CT and was more common with DK or DKA. These findings may help provide hepatic sampling recommendations and alter patient prognosis. Further research is needed to establish absolute and relative liver attenuation in dogs with correlation to histopathology and patient outcome.

1 | INTRODUCTION

In dogs, hepatic steatosis, or lipidosis, is a form of metabolic hepatopathy. It occurs when hepatocytes accumulate excessive fat due to disruptions in normal or abnormal metabolic pathways.^{1,2} This condition can arise from primary causes like familial hyperlipidemia or secondary to other diseases. In dogs, steatosis may be linked to endocrine disorders

like diabetes mellitus (DM), hypothyroidism, and hypercortisolism or toxic exposures such as aflatoxin. These lipid infiltrative vacuolar hepatopathies can be characterized as macrovesicular or microvesicular, with macrovesicular steatosis being more prevalent and characterized by cellular enlargement and internal cell derangements.¹ Traditionally, the diagnosis of this disease process requires cytological or histopathological confirmation.^{2–4} In people, hepatic steatosis may be diagnosed

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by measuring CT hepatic attenuation, which holds significant diagnostic value in evaluating liver donors.^{5–7} Similar studies have been performed in veterinary medicine in feline patients in experimental short-term fasting and groups at risk for hepatic lipidosis without a clear consensus on the value of CT in this disease, given variable appearance.^{8,9}

Decreased hepatic attenuation, measured in Hounsfield units (HU), indicates increased hepatic fat infiltration.^{5,6,10,11} Specific cutoff values for diagnosing hepatic steatosis in people include an absolute attenuation of less than 40 HU or a relative attenuation in which the liver is 10 HU or less than the spleen.^{7,12} A recent case report in the veterinary literature described negative (hypo) attenuation of the liver in a dog with DM secondary to hyperadrenocorticism, consistent with severe hepatic steatosis, as is reported in human literature.¹³ A similar case report in a feline patient also reported negative attenuation diffusely in the liver with a presumed diagnosis of hepatic lipidosis.¹⁴

DM is a known precipitant of hyperlipidemia due to the activation of counter-regulatory hormones, increased lipolysis, and gluconeogenesis. These alterations in lipid metabolism lead to hepatic lipidosis or steatosis, often accompanied by hepatomegaly in canine patients.^{3,15} At our institution, certain cases of diabetic ketoacidosis (DKA) have been associated with livers that appear hypoattenuating on CT scans, raising concerns about underlying hepatic steatosis. This observational insight prompts a broader evaluation of hepatic steatosis using CT attenuation values along with an analysis of associated hepatomegaly in dogs presenting with varying stages of DM.

However, the assessment of hepatic steatosis using CT attenuation and the degree of hepatomegaly associated with hypoattenuating liver have not been studied in dogs. Therefore, the purpose of this study is to describe hepatic parenchymal attenuation and CT-derived liver volume in dogs with various presentations of DM. We hypothesize that dogs diagnosed with diabetic ketosis (DK) or DKA will be more likely to have hypoattenuating livers compared with dogs diagnosed with DM without DK or DKA and that the degree of liver hypoattenuation will correlate with hepatomegaly.

2 | MATERIALS AND METHODS

2.1 | Experimental design and case selection

A retrospective search of medical records was conducted from January 1, 2015, to March 13, 2023, identifying dogs that underwent CT scans of the entire liver and had a clinical diagnosis of DM within 24 h of the CT scan. The clinical diagnosis of DM in this study was assigned based on the fulfillment of all three of the following criteria: (1) presence of appropriate clinical signs (polyphagia, polyuria, polydipsia, weight loss) and clinicopathologic abnormalities (hyperglycemia and glucosuria); (2) the inclusion of DM (with or without ketoacidosis) on the patient's master problem list; (3) medical record review by two internal medicine clinicians (one board-certified internal medicine specialist and one internal medicine resident) with consensus that a diagnosis of DM is appropriate.

Once diabetic dogs were identified, they were categorized into three groups based on clinical and biochemical parameters. Group 1 consisted of diabetic dogs without ketonuria, ketonemia, or metabolic acidosis. Group 2 included diabetic dogs with evidence of ketonuria or ketonemia but without metabolic acidosis based on blood gas analysis or evaluation of the anion gap. Group 3 comprised dogs with DKA, characterized by ketonuria and/or ketonemia, and metabolic acidosis based on blood gas analysis.

The collected data included, when available, patient identifiers (name, medical record number), date of presentation, demographic information (age, sex, breed, weight, body condition score), clinical presentation (including weight change and appetite), duration since initial DM diagnosis, ketonuria (Siemens Multistix 10 SG Reagent Strips, Siemens Healthineers), ketonemia (Nova Vet Ketone Meter, Nova Biomedical), acid/base status (Stat Profile Prime Plus, Nova Biomedical), prednisone and insulin administration, diabetic status classified by internal medicine clinicians, concurrent diagnosis and treatment of hypercortisolism, and liver cytology or histopathology findings at the time of the visit. Additional laboratory parameters included alanine aminotransferase (ALT), alkaline phosphatase (ALP), gamma-glutamyl transferase (GGT), total bilirubin (Tbili), albumin, and cholesterol levels.

Due to the retrospective study design, institutional animal care approval was not required.

2.2 | CT image analyses

All CT studies were performed using a 64-slice multidetector CT machine (GE LightSpeed VCT, GE Healthcare) with images acquired in the transverse plane using the following image acquisition parameters: helical scan mode, 100–120 kVp, 240–340 mA, slice thickness = 0.625–2.5 mm, tube rotation time = 1 s, pitch = 1, matrix = 512 × 512, and detail algorithm. All dogs were positioned in sternal recumbency under sedation or general anesthesia. The field of view required to include both the liver and spleen.

All CT images were retrieved in the DICOM format. A second-year diagnostic imaging resident performed region of interest (ROI) placements and measurements, which were reviewed by an ACVR board-certified veterinary radiologist using an open-source image analysis software (Osirix MD, Bernex).

The hepatic parenchyma was evaluated using noncontrast CT images. If the hepatic parenchyma attenuation (HPA) was hypoattenuating to the hepatic veins, it was defined as a visually hypoattenuating liver. The ROI selection protocol was created to include the majority of liver regions in all dogs while avoiding large vessels and ensuring representation of the various liver lobes. A total of six ROIs were selected per liver, four of which were aligned at the level of the left hepatic vein or just caudal, depending on the size of the liver, with one each at the following locations: one dorsal right, one ventral right, one dorsal left, and one ventral left. Two additional ROIs were selected at the level just cranial to the right kidney at the following locations: one in the caudate process of the caudate lobe and one in the caudoventral aspect of the mid to left liver (Figure 1). Similarly, the spleen was evaluated with ROI

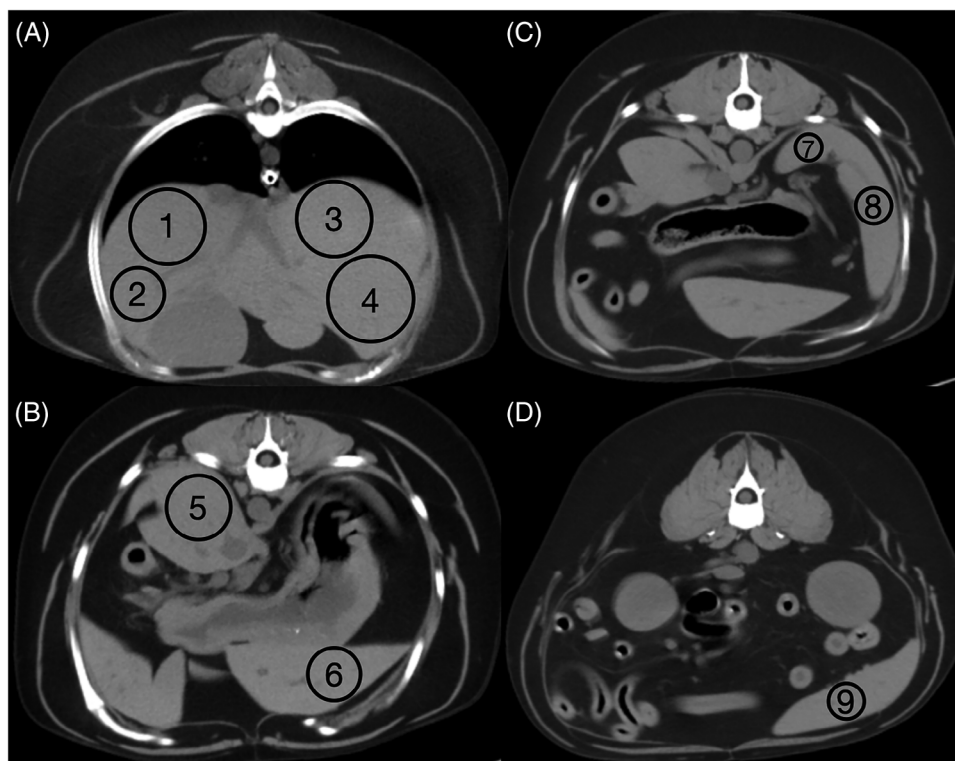


FIGURE 1 Placement of regions of interest (ROIs) for CT attenuation measurement in transverse CT images of the liver and spleen in dogs with diabetes mellitus. A, Transverse CT image at the level of the left hepatic vein showing ROIs in the liver: 1, dorsal right; 2, ventral right; 3, dorsal left; and 4, ventral left. B, Transverse CT image just cranial to the right kidney showing ROIs in the liver: 5, caudate process of the caudate lobe, and 6, caudoventral aspect of the mid to left liver. C, Transverse CT image showing ROIs in the spleen: 7, dorsal extremity, and 8, middle part. D, Transverse CT image showing ROI in the tail of the spleen (9). All ROIs were placed as large as possible, avoiding major vasculature and obvious nodules or masses.

measurements in three different anatomical locations: dorsal extremity, middle part, and ventral extremity, avoiding major vasculature and obvious nodules or masses (Figure 1). Means ($\pm$ standard deviation) of HPA and splenic parenchymal attenuation (SPA) were calculated separately for each dog. Based on the criteria used in people,^{7,12} canine livers on CT were defined as absolute hypoattenuation for mean HPA less than 40 HU^{7,16} or as relative hypoattenuation for mean HPA 10 HU less than the mean SPA¹² (Figure 2).

Collected CT data included the reason for CT examination, the date of CT acquisition, CT parameters, and quantitative measurements of liver and spleen HU as described previously; including mean, standard deviation (SD), and range of both liver and spleen in HU, presence of visually hypoattenuating liver compared with vasculature, presence of nodules or masses in the liver, additional CT hepatic abnormalities, and presence of adrenomegaly if available.

2.3 | Manual CT hepatic volumetry

CT hepatic volumetry was performed by a veterinarian who had received training supervised by a board-certified radiologist using a DICOM viewer (Horos 64-bit, version 3.3.6., Purview).^{17,18} The window width was set at 350 HU and the window level at 40 HU for all

dogs. The liver segmentation was performed by manually drawing the operator-defined ROI on precontrast transverse images of the entire liver from the cranial margin of the liver at the diaphragm to the most caudal margins of the liver adjacent to the right kidney and the spleen. The ROIs included the hepatic vessels within the liver parenchyma but excluded the gallbladder and visible hepatic lobe fissures and hepatic vessels that were present outside of the hepatic parenchymal margin. After manual drawing of the ROIs on the hepatic parenchyma with more than 20 slices, the CT-derived liver volume was computed using the following formula to estimate the liver volume: $\Sigma \{ \text{each slice area}(\text{cm}^2) \times \text{slice thickness}(\text{cm}) \} \times \text{total number of slices of hepatic parenchyma/number of slices}$.¹⁸

2.4 | Statistical analysis

Normality of the distribution of mean liver attenuation and liver volume for the entire population was assessed using the Shapiro–Wilk test. This test confirmed a normal distribution for liver volume across the entire population but indicated a nonparametric distribution for liver attenuation. Given the small sample size for each subgroup, the data for each group were treated as nonparametric. Consequently, descriptive statistics (mean and standard deviation) were calculated

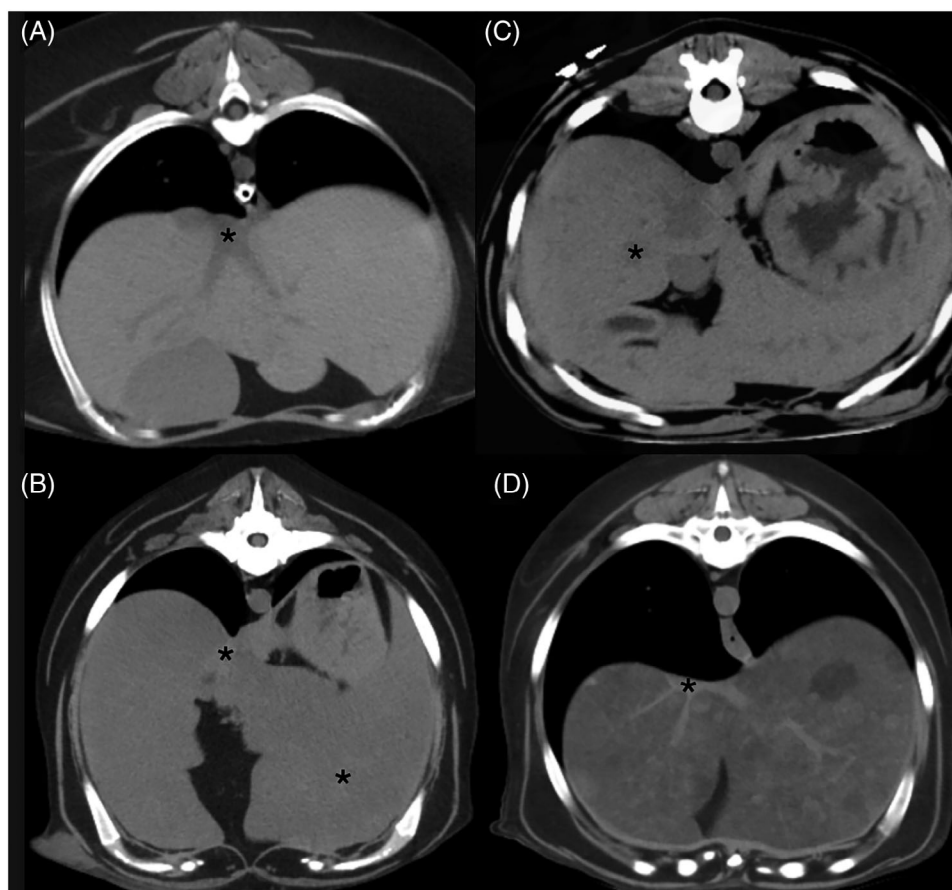


FIGURE 2 Transverse CT images of the liver at the level of the left hepatic vein in dogs with diabetes mellitus. A, Normal attenuation of the liver with a mean hepatic parenchymal attenuation (HPA) of 57 and a mean splenic parenchymal attenuation (SPA) of 51. B, Relative hypoattenuation of the liver with a mean HPA of 51 and a mean SPA of 68. C, Absolute hypoattenuation of the liver with a mean HPA of 26 and a mean SPA of 59. D, Absolute hypoattenuation of the liver with a mean HPA of -15 and a mean SPA of 50, with multifocal negatively attenuating nodules. Asterisks indicate either the caudal vena cava or the hepatic veins.

for liver volumes for the entire population. However, for each subgroup, the median and range were used to describe liver attenuation and liver volumes.

The Kruskal–Wallis test, followed by the post hoc Steel–Dwass test, was used to examine differences in absolute and relative liver attenuation and liver volumes between groups. Additionally, Spearman's rank correlation coefficient was performed to assess the relationship between mean liver attenuation and liver volume. A significance level of $P < .05$ was used to determine statistical significance.

3 | RESULTS

3.1 | Animals

3.1.1 | Case selection

Twenty-seven diabetic dogs were included in the study and divided into three groups. Fifteen dogs were in Group 1 (DM without ketoacidosis). Six dogs were in Group 2 (diabetic ketosis). Six dogs were

in Group 3 (diabetic ketoacidosis). General information, including age, body weight, and sex of the included cases, is described in Table 1.

A large variety of breeds were included, the most common being mixed breed dogs ($n = 4$), with no group having multiples of the same breed.

There were two dogs within Group 1 for which the time of initial DM diagnosis could not be determined from a medical record review. The median duration from DM diagnosis to the CT examination for all dogs was 6 months (range: 1 day–6 years). Group 1 ($N = 13/15$) had a median duration of 3 years (range: 0–6 years), Group 2 ($N = 6/6$) had a median of 1 year (range: 0–3 years), and Group 3 ($N = 6/6$) were diagnosed either at presentation or within the same week.

Five dogs had a confirmed concurrent diagnosis of hypercortisolism, four of which were receiving trilostane treatment. Of these four dogs, two were in Group 1 and considered well-controlled at the time of the CT study, based on the ACTH stimulation test. Among the three remaining dogs with poorly controlled hypercortisolism, one was untreated, and two were receiving trilostane. Two of these dogs were in Group 1, while one was in Group 3. Two of the dogs with a

TABLE 1 General information of included dogs with diabetes mellitus.

Median (range)	Group 1 (n = 15)	Group 2 (n = 6)	Group 3 (n = 6)	All (n = 27)
Age (years)	10 (6–14)	9 (6–12)	6 (2–8)	10 (2–14)
Weight (kg)	18.6 (6.8–37.9)	29.6 (9–56)	37.2 (31–51)	26.8 (5.8–56)
Sex and neuter status	MC: 7 FS: 8	MC: 2 FS: 2/Fl: 2	MC: 2/MI: 1 FS: 3	M: 12 F: 15
Body condition score	6/9 (3/9–7/9)	5/9 (3/9–9/9)	6.5/9 (4/9–8/9)	5/9 (3/9–9/9)

Abbreviations: MI, intact male; MC, male castrated; Fl, intact female; FS, female spayed.

TABLE 2 Biochemical analysis findings in dogs with diabetes mellitus.

	Group 1	Group 2	Group 3	Reference range
ALT	158 (27–804)	127	232 (105–455)	3–69 U/L
ALP	696 (95–2682)	2605	1389 (108–4500)	20–157 IU/L
GGT	19 (10–39)	14	27 (0–128)	5–16 IU/L
Tbili	0.4 (0.1–47.3)	3	0.6 (0.2–1.3)	0.10–0.80 mg/dL
Albumin	3.4 (2.9–6.6)	3	3 (2.3–4.2)	2.3–3.9 g/dL
Cholesterol	308 (239–846)	152	235 (173–411)	125–301 mg/dL

Note: The values are reported as median and range. Only 1 patient from Group 2 had biochemical analysis available.

confirmed diagnosis of hypercortisolism exhibited bilateral generalized adrenomegaly on CT, which is consistent with pituitary-dependent hypercortisolism. The remaining dogs did not show evidence of adrenomegaly or adrenal mass on CT.

Additionally, four dogs were receiving glucocorticoid therapy at the time of the CT examination. Two of these dogs had either been treated with a minimal dose of prednisolone (0.07 mg/kg/day) or had recently started treatment (1.1 mg/kg/day) 2 days prior to the CT study; both were in Group 1. The other two dogs had been on long-term prednisolone therapy for over a year (0.4 and 1.2 mg/kg/day, respectively) and were in Group 3.

At the time of the CT scan, one dog in Group 1, two dogs in Group 2, and four dogs in Group 3 were not receiving insulin therapy, while the remaining dogs were under insulin treatment. Serum glucose levels were measured for all dogs during the same visit as the CT scan. For dogs receiving insulin therapy, serum glucose levels varied due to differences in the timing of insulin administration and patient condition. Among the dogs not receiving insulin at the time of initial blood work, six had hyperglycemia, while one dog had normal serum glucose levels, likely due to the severity of the patient's illness.

3.1.2 | Biochemical information

Serum biochemical analysis was available for 17 of the 27 dogs, specifically for 11 of the 15 dogs in Group 1, one of the six dogs in Group 2, and five of the six dogs in Group 3. Hepatic enzyme levels, particularly ALT and ALP, were elevated across the groups. These findings are summarized in Table 2.

3.2 | CT attenuation of liver and spleen

In Group 1, one of 15 dogs had a relatively hypoattenuating liver, with a mean HPA of 17 HU below the mean SPA. The remaining dogs did not have absolute or relative hypoattenuating liver (Table 3). The dog with a relative hypoattenuating liver in Group 1 did not have hypercortisolism or exogenous glucocorticoid therapy. The median and range of HPA for Group 1 was 64 HU (range: 51–71 HU). Four dogs in this group were concurrently diagnosed with hypercortisolism, showing HPA values between 62 and 68 HU. Additionally, two dogs in Group 1, which had minimal or very short-term glucocorticoid exposure, exhibited HPAs of 53 and 70 HU, respectively.

In Group 2, four of six dogs had relative hypoattenuating livers (mean HPA 25, 15, 12, and 12 HU less than mean SPA). The remaining dogs did not have absolute or relative hypoattenuating livers (Table 3). None of these dogs were diagnosed with hypercortisolism or received glucocorticoid therapy. The median and range of HPA for Group 2 was 53 HU (range: 42–55 HU).

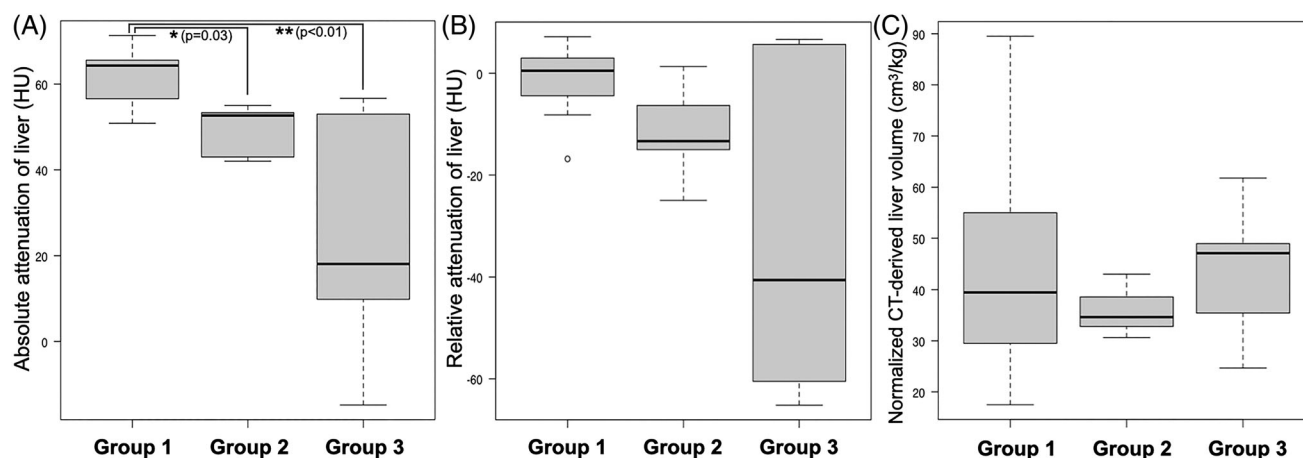
In Group 3, four of six dogs had absolute hypoattenuating livers, all of which were concurrently relative hypoattenuating (Table 3). Three of the four dogs showed visual hypoattenuation, characterized by the hepatic parenchyma being darker than the hepatic veins. One of the four dogs had a negative attenuating liver, consistent with fat. One patient with a hypoattenuating liver (25 HU) was concurrently diagnosed with hypercortisolism. The median and range of HPA for Group 3 was 18 HU (range: –15 to 57 HU). Additionally, one dog had a concurrent hypercortisolism diagnosis with an HPA of 26 HU, while two dogs that had received long-term glucocorticoid therapy exhibited HPAs of 53 and 57 HU, respectively.

TABLE 3 Number of relative and absolute hypoattenuation of the liver and median liver volume in dogs with diabetes mellitus in each group.

Group	Relative hypoattenuation (HPA < SPA–10 HU)	Absolute hypoattenuation (HPA < 40 HU)	Hepatic volume median (range; cm ³ /kg)
1	1/15 (7%)	0	39.5 (17.5–89.5)
2	4/6 (67%)	0	34.7 (30.6–43.0)
3	4/6 (67%)	4/6 (67%)	47.1 (24.7–61.8)

Note: The table shows the number of patients and the percentage of cases with relative or absolute hypoattenuation. The third column shows the median liver volumes and ranges for each group.

Abbreviations: HPA, hepatic parenchymal attenuation. SPA, splenic parenchymal attenuation.

**FIGURE 3** Comparison of liver attenuation and volume among different groups of dogs with diabetes mellitus. Box plots display the median, interquartile range, and outliers. A, Absolute attenuation of the liver (HU) in groups 1, 2, and 3. Statistically significant differences are indicated by * $P < .03$ and ** $P < .01$. B, Relative attenuation of the liver (HU) in groups 1, 2, and 3. No statistical differences are present among groups ($P = .07$). C, Normalized CT-derived liver volume (cm³/kg) in groups 1, 2, and 3. No statistical differences are present among groups ($P = .32$).

Statistical analysis revealed that having absolute liver hypoattenuation differed between groups ($P < .01$). Post hoc evaluation indicated a significant difference in absolute liver hypoattenuation between Group 1 and Group 2 ($P = .03$) as well as between Group 1 and Group 3 ($P < .01$, Figure 3A), but not between Group 2 and Group 3. Having relative liver hypoattenuation did not differ between groups ($P = .07$, Figure 3B).

Nonspecific nodular change was present in 5 of 15 in Group 1, 3 of 6 in Group 2, and 1 of 6 in Group 3. Additional significant abnormalities on CT included a solitary liver mass ($N = 1$, Group 1) and hepatic abscess ($N = 1$, Group 1).

3.3 | Manual CT hepatic volumetry

The normalized CT-derived liver volumes across all studied diabetic dogs were calculated, revealing a mean ($\pm$ standard deviation) liver volume of 42.9 (± 17.3) cm³/kg. By group, the median liver volumes were as follows: 39.5 (range: 17.5–89.5) cm³/kg in Group 1, 34.7 (30.6–43.0) cm³/kg in Group 2, and 47.1 (24.7–61.8) cm³/kg in Group 3 (Table 3). When categorized by CT liver attenuation, the median liver volumes were 43.3 (17.5–89.5) cm³/kg for dogs with normal liver

attenuation, 37.0 (24.7–61.8) cm³/kg for those with relative hypoattenuation, and 35.4 (24.7–61.8) cm³/kg for those exhibiting absolute hypoattenuation.

No significant differences were identified in liver volumes between the groups ($P = .46$, Figure 3C). Additionally, no correlation was found between mean liver attenuation and liver volume ($P = .32$).

3.4 | Cytological and histological analyses

Nine of 27 dogs had cytology ($N = 6$) or histopathology ($N = 3$) performed of the liver, which revealed normal liver ($N = 1$), lipid accumulation ($N = 5$), glycogen accumulation ($N = 2$), nodular hyperplasia ($N = 2$), and a solitary mass diagnosed as hepatocellular carcinoma ($N = 1$).

In Group 1, two dogs had cytologic evidence of lipid infiltration, neither of which were the dogs with relative hypoattenuation. In Group 2, one of the dogs with a relative hypoattenuating liver had a diagnosis of lipid infiltration on cytology. In Group 3, the negative HPA dog had a histopathologic diagnosis of severe hepatic lipidosis from a necropsy. One dog with a visually hypoattenuating liver had cytology consistent with lipid accumulation and vacuolar hepatopathy.

4 | DISCUSSION

This study confirmed that hypoattenuation of the liver may be present in dogs with DM. As hypothesized, dogs with DK or DKA had lower hepatic attenuation compared with those without these conditions. However, contrary to our expectations, there were no significant differences in liver volume between the groups or when compared with the degree of liver attenuation. This suggests that while liver attenuation decreases with the severity of diabetes, hepatomegaly does not differ in diabetic dogs, regardless of the presence of ketoacidosis. These findings highlight the complexity of hepatic changes in diabetic dogs. In the present study, all patients in the DKA group and the majority of the remaining patients who had blood work to assess liver enzymes exhibited elevated hepatic enzyme activities, which is common with DM.^{2,3,19} Liver function parameters were not routinely measured in this study, likely due to the fact that hepatic dysfunction is rare in canine diabetes. Hepatic lipidosis in dogs can be suspected from cytology samples, but a definitive diagnosis typically requires histology. However, this can be challenging to perform in a critically ill patient. Differentiating steatosis between other types of vacuolar hepatopathies, such as those involving glycogen, requires histological analysis using special stains like periodic acid–Schiff for glycogen and oil red O for fat. These distinctions are essential for understanding the specific physiological processes leading to steatosis and its potential impact on liver function. As the lipid vacuolization of hepatocytes worsens, the cells can become swollen, which contributes to the hepatomegaly that is common in canine diabetes. If hepatocyte swelling is severe and chronic, it can cause hepatocyte death, fibrosis, and even cirrhosis.^{20,21}

While hepatic steatosis has been diagnosed by measuring CT hepatic attenuation in humans,^{5–7} CT appearance of this condition in dogs has only been documented as negative attenuation in one with DM secondary to hyperadrenocorticism.¹³ Our study found that liver attenuation in dogs with DKA or DK was significantly lower compared with the dogs without DK or DKA. Notably, dogs presenting with DKA had the most hypoattenuating livers, the only absolute hypoattenuating livers, and the only negative attenuating liver. The difference between liver and spleen attenuation was also larger in the DKA group compared with the DK group, suggesting more severe hypoattenuation in dogs with more severe diabetic states. However, the relative attenuation of the liver was not significantly different among varying severities of diabetic status.

Endocrine disorders, such as hypercortisolism or hypothyroidism, as well as exogenous glucocorticoid treatment, are known to affect glycogen and fat deposition in the liver.^{22,23} Glycogen has a CT attenuation range of 50–70 HU and is known to increase liver attenuation in humans.²⁴ In dogs, hyperadrenocorticism is associated with increased HPA due to glycogen deposition,^{25,26,27} though it can also decrease HPA due to hepatic steatosis, particularly in cases with concurrent DM.¹³ It is important to note that one dog each from the DKA and DK groups had concurrent hypercortisolism and hypothyroidism with absolute and relative liver hypoattenuation, respectively. Two dogs in

Group 3, which were on long-term glucocorticoid therapy, showed HPA values similar to those in dogs without liver disease, with values of 53 and 57 HU, while four other dogs in Group 3 exhibited absolute liver hypoattenuation. The absence of liver hypoattenuation in these two dogs could be attributed to glycogen deposition from long-term glucocorticoid therapy. Since concurrent endocrine disorders may have contributed to the liver attenuation in both directions by increasing or decreasing it, in addition to DM, assessing the true extent of steatosis in these dogs based solely on CT liver attenuation was challenging. No other comorbidities were identified in patients with hypoattenuating livers.

In humans, progressive hepatic fat infiltration significantly alters liver attenuation, with different levels of fat infiltration considered clinically important.^{6,7,12,28,29} This relationship may be similarly important in dogs, although criteria for evaluating hepatic steatosis or lipidosis by CT in dogs have not been established. In humans, fatty infiltration of the liver is diagnosed when hepatic attenuation falls below 48 HU,⁷ with attenuation at 40 HU correlating with approximately 30% fatty change.¹⁶ Since normal liver CT values have been reported as 50–65 HU in humans¹⁶ and 60–70 HU in dogs,³⁰ a cutoff of less than 40 HU to diagnose absolute liver hypoattenuation is stringent. This criterion is likely to identify dogs with significant hepatic steatosis but may overlook cases of mild steatosis. To address this, we also reported the relative attenuation of the liver compared with the spleen. Indeed, in humans, the liver is considered relatively hypoattenuating when the difference between SPA and HPA (SPA-HPA) exceeds 10 HU.¹² Normal SPA-HPA differences have been reported as ranging from –1 to –18 HU in humans,³¹ and a mean of 4.7 (± 7.8) HU in dogs.³² The presence of a relatively hypoattenuating liver in our study may reflect normal variation but is likely to include all dogs with mild steatosis. The absence of established reference points also could lead to misdiagnosis of relative liver hypoattenuation, particularly due to diffuse splenic attenuation changes. Absolute hypoattenuation of the liver, which was only identified in dogs with DKA in our study, is considered a more significant and reliable indicator of hepatic steatosis in humans. Despite these findings, the impact of hypoattenuating liver on patient outcomes remains uncertain. In one notable case, a patient with negative liver attenuation was euthanized due to poor prognosis and lack of response to treatment, with severe hepatic steatosis later confirmed by histopathology, suggesting that liver disease was a significant contributor to patient morbidity.

In the present study, mean normalized liver volume was higher in diabetic dogs compared with previously reported ranges in dogs without liver disease.¹⁷ This indicates that hepatomegaly is a common change in diabetic dogs. However, liver volume did not differ significantly among the diabetic groups nor did it correlate with the degree of hypoattenuation, suggesting that hepatomegaly in diabetic dogs is not solely due to hepatic steatosis. The metabolic derangements of DM can lead to glycogen accumulation in the hepatocytes, which can cause vacuolation and swelling of the hepatocytes with hepatomegaly, as was previously described with lipid vacuolation. These alterations in hepatocellular storage and subsequent hepatomegaly are considered

common in canine diabetes, meaning that liver volume may not be a helpful tool for estimating the degree of lipid accumulation, specifically.

Limitations of this study include its retrospective nature with variable diagnoses, limited clinicopathologic evaluations, variable treatments, limited follow-up, and lack of a liver phantom CT study. This study may have underestimated patients with concurrent disease processes, such as hypercortisolism, that may affect the evaluation of the hepatic parenchyma, given that these are not routinely tested for, especially in sick patients. The CT examinations were not standardized due to variable acquisition times and patient protocols, although the same CT scanner and base protocols were used. Since CT parameters can influence the actual CT attenuation values, a liver phantom should ideally have been used in this study for calibration. Ideally, more patients would be enrolled with better classification of DM status and following established HU ranges for normal livers while investigating the variation of lipid infiltration via histopathology, measuring triglyceride levels, and correlating these findings to clinical relevance and patient outcome. Follow-up imaging for evaluation of changes to the liver with clinical treatment would also give more insight into changes in HPA over time in diabetic patients.

5 | CONCLUSION

Hypoattenuation of the liver can be seen in dogs with DK or DKA and is more severe when compared with dogs without acidosis or ketoacidosis. This hypoattenuation is most likely a result of hepatic steatosis, suggesting that this fatty infiltration is more severe in dogs with DKA. Identification of relative or absolute hypoattenuation of the liver in dogs should prompt evaluation of glycemic control. Further evaluation of liver CT in dogs with DK or DKA will be helpful in determining if this modality can aid in guiding diagnostics or influencing prognosis. Further research in veterinary patients is needed to establish other potential disease processes that may cause similar liver hypoattenuation, in addition to standardizing liver attenuation values with correlation to histopathology, therapies, and patient outcomes.

LIST OF AUTHOR CONTRIBUTIONS

Category 1

- (a) Conception and Design: Buckley, Fulkerson, Murakami
- (b) Acquisition of data: Buckley, Fulkerson, Derre, Woolcock, Murakami
- (c) Analysis and Interpretation of Data: Buckley, Fulkerson, Derre, Woolcock, Murakami

Category 2

- (a) Drafting the Article: Buckley, Fulkerson, Murakami
- (b) Revising the Article: Buckley, Fulkerson, Derre, Woolcock, Murakami

Category 3

- (a) Final Approval: Buckley, Fulkerson, Derre, Woolcock, Murakami

Category 4

- (a) Agreement to be accountable for all aspects of the work ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved: Buckley, Fulkerson, Derre, Woolcock, Murakami

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

PREVIOUS PRESENTATION DISCLOSURE

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EQUATOR NETWORK DISCLOSURE

An EQUATOR network checklist was not used.

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REFERENCES

1. Wilcock B. Chapter 29: Histopathology. In Day MJ, *Washabau RJ Canine and Feline Gastroenterology*. Elsevier; 2013;333-385.
2. Nelson RW, Couto CG. Part 4: hepatobiliary and exocrine pancreatic disorders. *Small Animal Internal Medicine* 6ed Elsevier. 2019.
3. Ettinger SJ, Feldman EC, Côté E. Textbook of Veterinary Internal Medicine: Diseases of the Dog and the Cat. Elsevier; 2017.
4. Hunt GB, Luff JA, Daniel L, Bergh RVanDen. Evaluation of hepatic steatosis in dogs with congenital portosystemic shunts using oil red O staining. *Vet Pathol*. 2013;50(6):1109-1115. doi:10.1177/0300985813481609
5. Hahn L, Reeder SB, Del Rio AM, Pickhardt PJ. Longitudinal changes in liver fat content in asymptomatic adults: hepatic attenuation on unenhanced ct as an imaging biomarker for steatosis. *AJR Am J Roentgenol*. 2015;205(6):1167-1172. doi:10.2214/AJR.15.14724
6. Zheng D, Tian W, Zheng Z, Gu J, Guo Z, He X. Accuracy of computed tomography for detecting hepatic steatosis in donors for liver transplantation: a meta-analysis. *Clin Transplant*. 2017;31(8):e13013-n/a. doi:10.1111/ctr.13013
7. Yajima Y, Narui T, Ishii M, et al. Computed tomography in the diagnosis of fatty liver: total lipid content and computed tomography number. *Tohoku J Exp Med*. 1982;136(3):337-342. doi:10.1620/tjem.136.337
8. Nakamura M, Chen HM, Momoi Y, Iwasaki T. Clinical application of computed tomography for the diagnosis of feline hepatic lipidosis. *J Vet Med Sci*. 2005 Nov;67(11):1163-1165. doi:10.1292/jvms.67.1163
9. Lam R, Niessen SJ, Lamb CR. X-Ray attenuation of the liver and kidney in cats considered at varying risk of hepatic lipidosis. *Vet Radiol Ultrasound*. 2014;55(2):141-146. doi:10.1111/vru.12113
10. Schwenzer NF, Springer F, Schraml C, Stefan N, Machann J, Schick F. Non-invasive assessment and quantification of liver steatosis by ultrasound, computed tomography and magnetic resonance. *J Hepatol*. 2009;51(3):433-445. doi:10.1016/j.jhep.2009.05.023
11. Rogier Julien, Roulet Stéphanie, Cornélis François, et al. Noninvasive assessment of macrovesicular liver steatosis in cadaveric donors based on computed tomography liver-to-spleen attenuation ratio. *Liver Transpl*. 2015;21(5):690-695. doi:10.1002/lt.24105

12. Limanond P, Raman SS, Lassman C, et al. Macrovesicular hepatic steatosis in living related liver donors: correlation between CT and histologic findings. *Radiology*. 2004 Jan;230(1):276-280. doi:[10.1148/radiol.2301021176](https://doi.org/10.1148/radiol.2301021176)
13. Carloni Andrea, Paninarova Michaela, Cavina Damiano, et al. Negative hepatic computed tomographic attenuation pattern in a dog with vacuolar hepatopathy and hepatic fat accumulation secondary to Cushing's syndrome. *Vet Radiol Ultrasound*. 2019;60(5):E54-E57. doi:[10.1111/vru.12568](https://doi.org/10.1111/vru.12568)
14. Heo SH, Yoon YM, Hwang TS, Jung DI, Lee HC. Imaging diagnosis of hepatic lipidosis in a cat. *Korean J Vet Res*. 2018;58(2):99-101. doi:[10.14405/kjvr.2018.58.2.99](https://doi.org/10.14405/kjvr.2018.58.2.99)
15. Xenoulis PG, Steiner JM. Lipid metabolism and hyperlipidemia in dogs. *Vet J*. 2010;183(1):12-21. doi:[10.1016/j.tvjl.2008.10.011](https://doi.org/10.1016/j.tvjl.2008.10.011)
16. Kodama Yoshihisa, Ng ChaanS, Wu TsungT, et al. Comparison of CT methods for determining the fat content of the liver. *AJR Am J Roentgenol*. 2007;188(5):1307-1312. doi:[10.2214/AJR.06.0992](https://doi.org/10.2214/AJR.06.0992)
17. Kinoshita K, Moore G, Murakami M. Body weight as a preferred method for normalizing the computed tomography-derived liver volume in dogs without hepatic disease. *Vet Sci*. 2024;11(4):153. doi:[10.3390/vetsci11040153](https://doi.org/10.3390/vetsci11040153)
18. Kinoshita K, Kurihara H, Moore GE, Murakami M. Pilot study: the effects of slice parameters and the interobserver measurement variability in computed tomographic hepatic volumetry in dogs without hepatic disease. *Vet Sci*. 2023;10(3):. doi:[10.3390/vetsci10030177](https://doi.org/10.3390/vetsci10030177)
19. Sepesy LM, Center SA, Randolph JF, Warner KL, Erb HN. Vacuolar hepatopathy in dogs: 336 cases (1993-2005). *J Am Vet Med Assoc*. 2006;229(2):246-252. doi:[10.2460/javma.229.2.246](https://doi.org/10.2460/javma.229.2.246)
20. Winkle TV, Cullen JM, van den Ingh TedSGAM, Charles JA, Desmet VJ. Chapter 8: morphological classification of parenchymal disorders of the canine and feline liver: 3. Hepatic abscesses and granulomas, hepatic metabolic storage disorders and miscellaneous conditions. In: Rothuizen J, ed. *WSAVA Standards for Clinical and Histological Diagnosis of Canine and Feline Liver Diseases*. W.B. Saunders; 2006;103-116.
21. Webster CynthiaRL, Center SharonA, Cullen JohnM, et al. ACVIM consensus statement on the diagnosis and treatment of chronic hepatitis in dogs. *J Vet Intern Med*. 2019;33(3):1173-1200. doi:[10.1111/jvim.15467](https://doi.org/10.1111/jvim.15467)
22. Tanase DanielaMaria, Gosav EvelinaMaria, Neculae Ecaterina, et al. Hypothyroidism-induced nonalcoholic fatty liver disease (HIN): mechanisms and emerging therapeutic options. *Int J Mol Sci*. 2020;21(16):5927.
23. In Johnston AN. Hepatobiliary Disease. *Clinical Medicine of the Dog and Cat*. CRC Press; 2022;360-377.
24. Doppman JohnL, Cornblath Marvin, Dwyer AndrewJ, Adams AnthonyJ, Girtton MaryE, Sidbury James. Computed tomography of the liver and kidneys in glycogen storage disease. *J Comput Assist Tomogr*. 1982;6(1):67-71.
25. Manfredi M, De Zani D, Oldani E, et al. Computed tomography (CT) evaluation of hepatic and bone density in dogs with and without hyperadrenocorticism. *Vet Radiol Ultrasound*. 2020;61(1):102. Abstract.
26. Costa LAVS, Oliveira DC, Lopes BF, Lanis AB, Teixeira MW, Costa FS. Quantitative computed tomography of the liver in dogs submitted to prednisone therapy. *Arq Bras Med Vet Zootec*. 2013;65:1084-1090.
27. Oliveira DC, Costa LAVS, Lopes BF, et al. Computed tomography in the diagnosis of steroidal hepatopathy in a dog: case report. *Arq Bras Med Vet Zootec*. 2011;63:36-39.
28. Jawahar A, Gonzalez B, Balasubramanian N, Adams W, Goldberg A. Comparison of correlations between lipid profile and different computed tomography fatty liver criteria in the setting of incidentally noted fatty liver on computed tomography examinations. *Eur J Gastroenterol Hepatol*. 2017;29(12):1389-1396. doi:[10.1097/MEG.0000000000000972](https://doi.org/10.1097/MEG.0000000000000972)
29. Pamilo M, Sotaniemi EA, Suramo I, Lähde S, Arranto AJ. Evaluation of liver steatotic and fibrous content by computerized tomography and ultrasound. *Scand J Gastroenterol*. 1983;18(6):743-747. doi:[10.3109/00365528309182089](https://doi.org/10.3109/00365528309182089)
30. Rossi F, Morandi F, Schwarz T. Liver, gallbladder and spleen. In: Schwarz T, Saunders J, eds. *Veterinary Computed Tomography*. Wiley-Blackwell; 2011;:297-314.
31. Park YS, Park SH, Lee S, et al. Biopsy-proven nonsteatotic liver in adults: estimation of reference range for difference in attenuation between the liver and the spleen at nonenhanced CT. *Radiology*. 2011;258(3):760-766. doi:[10.1148/radiol.10101233](https://doi.org/10.1148/radiol.10101233)
32. Costa LAVS, Maestri LFDP, Maia JA, et al. Hepatic radiodensity in healthy dogs by helical computed tomography/Radiodensidade hepática de cães hígidos por tomografia computadorizada helicoidal. *Ciencia Rural*. 2010;40(4):888-893. doi:[10.1590/S0103-84782010005000053](https://doi.org/10.1590/S0103-84782010005000053)

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